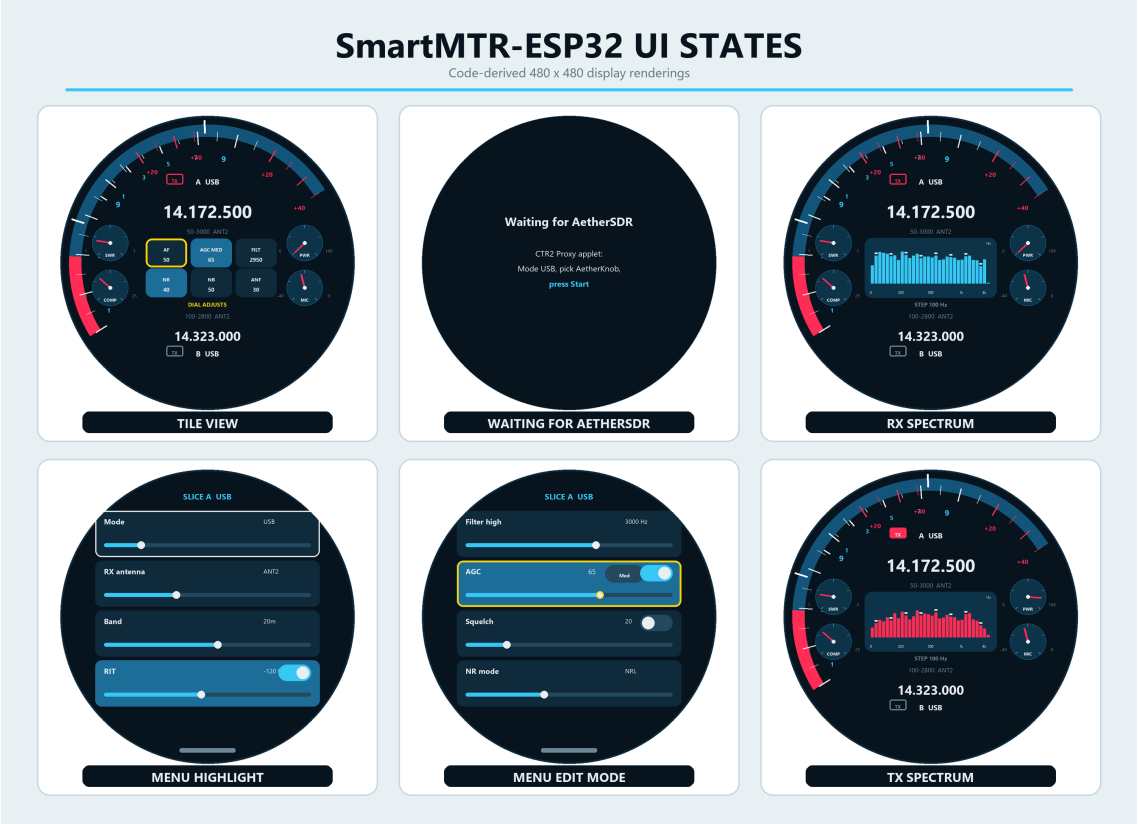
SmartMTR-ESP32 User Manual

Practical guide for AetherSDR and FlexRadio



Source-based documentation - Repository version 2026-10-23

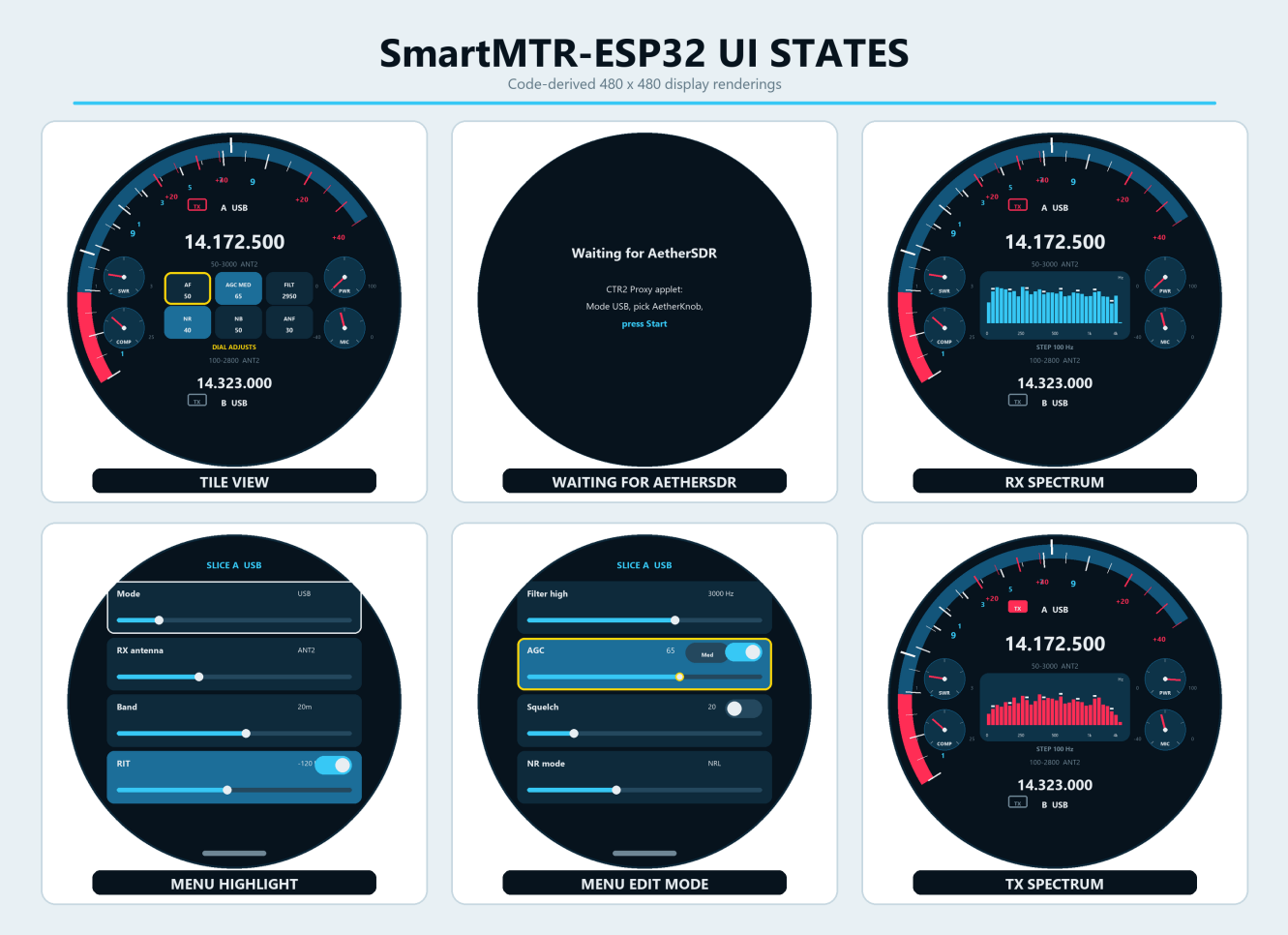
This manual describes the current SmartMTR-ESP32 device, also called “AetherKnob.” It is for radio amateurs who want to use the controller with AetherSDR and a FlexRadio. It is an operating manual, not a development guide.

Scope and verification status

The control logic, SmartSDR protocol handling, menu commands, meter movement, CTR2 data handling, and spectrum helpers are implemented and covered by offline tests. A desktop preview with scripted touch and dial actions is also included in the project.

The following items have not yet been proven on the real target hardware: display startup, touch orientation, encoder counts per detent, and UDP meter data carried through CTR2. This manual marks these areas as implemented but not yet hardware-verified. A successful firmware build or desktop preview does not replace a test on the real CrowPanel.

UI overview



Source-based UI renderings, not hardware photographs.

1. Device overview

SmartMTR-ESP32 is a display and control unit for a FlexRadio station. It runs on the Elecrow CrowPanel 2.1\" HMI ESP32 Rotary Display (DHE03921D). The board has an ESP32-S3, a round 480x480 IPS touch screen, and a rotary dial that can also be pressed. The round display design is based on SmartMTR by Dariusz Socha, SQ5X.

The controller does not connect directly to the radio by Wi-Fi or Ethernet. It connects by USB to the PC running AetherSDR. The CTR2 Proxy applet in AetherSDR carries status information, meter data, and control commands between AetherKnob and the FlexRadio. AetherKnob works as a separate SmartSDR API client and binds to the active AetherSDR or SmartSDR GUI station.

2. Connections and controls

Item	Purpose
Round display	Shows up to two slices, meters, spectrum or tiles, connection status, and the menu.
Touch screen	Supports tap, touch-and-hold, and swipe gestures.
Rotary dial	Tunes the radio, changes values, and moves through the menu. One mechanical detent is intended to produce one control step, but this is not yet hardware-verified.
Dial push button	Supports a short click and a press-and-hold of at least 0.6 seconds.
USB-5V-IN (4-pin MX1.25)	Supplies power and provides the native USB connection to the PC through the supplied MX1.25-to-USB cable. It is also the flashing port.
UART (4 pins: RX, TX, 5V, GND)	Provides firmware logs at 115200 baud. Use a 3.3 V USB-to-serial adapter and do not connect its power output to the board's 5 V pin.
I2C (4 pins: SCL, SDA, 5V, GND)	Spare expansion connector. The same bus already contains the touch controller at address 0x15 and the PCF8574 at 0x21.
RESET	Restarts the device. Used with BOOT to enter the ROM download mode.
BOOT	Hold this button while resetting to enter flashing mode.

When the firmware is running, the native USB port is a vendor-specific HID device. It does not provide the normal USB serial console. Firmware logs are available on the UART connector instead.

3. Powering up and connecting

1. Switch on the FlexRadio and start AetherSDR on the PC.
2. Connect AetherSDR to the required FlexRadio station.
3. Connect AetherKnob to the PC through USB-5V-IN.
4. Open the CTR2 Proxy applet in AetherSDR.
5. Set Mode to USB.
6. Select AetherKnob, shown by USB as AETHER_KNOB.
7. Press Start.

Expected display states:

Display	Meaning
AetherKnob - Plug into the PC running AetherSDR	No USB host is connected, or the USB connection has been removed.
Waiting for AetherSDR with CTR2 Proxy instructions	USB is connected, but CTR2 Proxy has not started the link.
Radio connected - Waiting for a SmartSDR or AetherSDR station	The radio is reachable through the proxy, but no suitable GUI station has been bound yet.
Round meter face with no status overlay	AetherKnob is bound to a station and ready to use.

The startup and binding behavior is implemented and offline-tested. The full chain with the real panel, AetherSDR, and a FlexRadio has not yet been recorded as a completed hardware test in this project.

4. Main display

The top slice area shows the active slice of the bound station. If the station does not report an active slice, AetherKnob uses the first slice that belongs to that station. A second slice from the same station is shown at the bottom when one is available.

Display item	Meaning
Large frequency	Slice frequency in **MHz.kHz.Hz** format.
Slice letter and mode	For example, **A USB**.
Filter and antenna line	Lower and upper filter edges, followed by the RX antenna.
S-meter arcs	Receive level for each slice, linear in dB from S0 to S9+40. The white marker shows the held peak.
SWR	Standing wave ratio, scale 1 to 3.
COMP	Compression level, scale 0 to 25 dB.
PWR	Forward power, scale 0 to 100 W.
MIC	Microphone level, scale −40 to 0 dB.
TX beside a slice	Shows and selects the transmit slice. See section 11.
STEP … Hz	Current tuning step.
DIAL ADJUSTS	The dial is changing a selected parameter instead of tuning.
LOCKED	Tuning is locked for the active slice.
Center area	Audio spectrum or six quick-control tiles.

5. Using the rotary dial

On the main display

Action	Result
Turn the dial	Tunes the active slice by one tuning step for each detected detent. Turning has no tuning effect while the slice is locked.
Short click	Selects the next tuning step: 10, 50, 100, 500, 1000, 5000, 10000 Hz, then back to 10 Hz.
Hold for at least 0.6 seconds	Turns Tune lock on or off for the active slice.
Turn while a tile has a yellow outline	Changes the selected tile parameter instead of frequency.
Short click in parameter mode	Leaves parameter mode, so the dial tunes again.

Tapping the same yellow-outlined tile also returns the dial to tuning. Encoder direction and the number of electrical counts per mechanical detent are set in the firmware but have not yet been confirmed on the real panel.

In the menu

When no row has a yellow edit outline, turning moves the white highlight. Clicking the dial starts editing the highlighted row. Turning then changes its value. See section 8 for full menu operation.

6. Touch operation

Gesture	Result on the main display
Tap a tile	Assigns that parameter to the rotary dial and gives the tile a yellow outline. Tap it again to return the dial to tuning.
Touch and hold a tile	Runs the tile's quick action. See section 7.
Tap a TX label	Makes that slice the transmit slice. The slice must belong to the bound station.
Swipe down	Opens the menu when the device is bound to a station.
Swipe left or right	Switches between spectrum and tiles. If no current spectrum is available, the tiles remain visible.
Tap the spectrum	Shows the tiles.

In the menu, swipe vertically to scroll the list. A vertical swipe that begins on a slider can still take over as list scrolling. Drag a slider sideways to change it. Tap switches and the AGC mode button.

Touch coordinates and gestures are implemented and can be exercised in the desktop preview, but touch orientation is not yet verified on the CrowPanel.

7. Tiles and quick controls

The value shown on each tile belongs to the active slice.

Tile	Tap, then turn the dial	Touch and hold
AF	Changes audio level from 0 to 100	Steps the AGC mode: off → slow → med → fast → off. AF has no separate on/off switch.
AGC	Changes AGC threshold from 0 to 100. When AGC is off, it changes fixed receive gain instead.	Steps the AGC mode.
FILT	Changes filter width symmetrically around the passband center in 50 Hz steps, limited to 50-12000 Hz.	Selects the next tuning step.
NR	Changes NR level from 0 to 100.	Turns NR on or off.
NB	Changes NB level from 0 to 100.	Turns NB on or off.
ANF	Changes ANF level from 0 to 100.	Turns ANF on or off.

An enabled function has a brighter tile background. A yellow outline means only that the tile's parameter is currently assigned to the rotary dial.

8. Menu

Opening, closing, and scrolling

- Swipe down on the bound main display to open the menu.
- Swipe up or down in the list to scroll.
- Tap the grip at the bottom, or swipe upward from it, to close the menu.
- You can also hold the dial when no row is being edited.
- The menu closes after 8 seconds without touch or dial activity.
- It also closes if the station binding is lost.

Closing the menu normally confirms any pending list choice. Holding the dial while editing does something different: it leaves edit mode and discards a list choice that has not yet been sent.

Editing by touch

- Tap a value or choice row to assign it to the dial.
- Tap the same row again to leave edit mode.
- Drag a slider sideways to change a level. Level changes are sent while you

drag. A list choice is shown in yellow and sent when you lift your finger.

- Tap a switch to turn a function on or off.
- Tap the small AGC mode button to step through **Off**, **Slow**, **Med**, and

Fast.

- On a combined level-and-switch row, touching and holding the row also toggles

its on/off setting.

Editing with the rotary dial

- White outline: the row is highlighted, but not being edited. Turn the

dial to move through the list. The list scrolls to keep the row visible.

• Yellow outline: the row is being edited. Turn the dial to change its

value or choice. For a plain on/off row, clockwise means on and anticlockwise means off.

• Short-click a white row to start editing it.

• On a combined level-and-switch row, a short click while editing toggles the

switch and leaves the level assigned to the dial.

• On the AGC row, a short click while editing steps the AGC mode and leaves

the level assigned to the dial.

• On other rows, a short click ends editing and immediately sends a previewed

choice. Without a click, a dial-selected choice is sent after the dial has been still for 0.7 seconds.

• Hold the dial while editing to go back and discard a choice that has not yet

been sent.

Yellow value text is a list choice that has not yet been finally sent. A brighter row background means the function is enabled; it is separate from the yellow edit outline.

9. Menu items

The names below are exactly the names shown on the device.

Receive

Item	Purpose and range
Mode	Selects a mode from the list supplied by the radio.
RX antenna	Selects a receive antenna from the list supplied by the radio.
Band	Selects 160 m, 80 m, 60 m, 40 m, 30 m, 20 m, 17 m, 15 m, 12 m, 10 m, 6 m, WWV, or GEN. The radio may recall a band stack and recreate the slice.
RIT	On/off and −9990 to +9990 Hz in 10 Hz steps.
Filter low	Lower filter edge from −12000 to +12000 Hz in 50 Hz steps. Hidden in FM and NFM.
Filter high	Upper filter edge from −12000 to +12000 Hz in 50 Hz steps. Hidden in FM and NFM. Minimum filter width is 50 Hz; allowed limits also depend on mode and, for CW, pitch.
AGC	Off/Slow/Med/Fast mode and level 0-100. With AGC off, the level is fixed receive gain. Otherwise it is the AGC threshold.
Squelch	On/off and threshold 0-100.
NR mode	Off, NR, NRL, NRS, RNN, or NRF. NRL/NRS/RNN/NRF are FLEX-8000/Aurora functions and are ignored by a FLEX-6000.
NR amount	Level 0-100 for the active NR type, where that type has an adjustable level.
ANF mode	Off, ANF, ANFL, or ANFT. ANFL/ANFT are FLEX-8000/Aurora functions and are ignored by a FLEX-6000.
ANF amount	Level 0-100 for the active ANF type, where that type has an adjustable level.
WNB	Wide Noise Blanker on/off and level 0-100.
APF	Audio Peak Filter on/off and level 0-100. Visible only in CW modes.
Mute	Turns audio for the active slice on or off.
Pan	Stereo position 0-100 in steps of two. The display runs from **L100** through **L2**, then **C**, then **R2** through **R100**.
Diversity	Turns diversity on or off for the active slice.
DAX channel	Off or channel 1-8.
Tune lock	Locks or unlocks frequency tuning for the active slice.

Transmit

Item	Purpose and range
RF power	Transmit power 0-100.
Tune power	TUNE power 0-100. This sets only the power value; it does not start TUNE.
Mic gain	Microphone gain 0-100.
Compander	On/off and level 0-100.
Monitor	Transmit monitor on/off and level 0-100.
TX antenna	Selects a transmit antenna from the list supplied by the radio.
XIT	On/off and −9990 to +9990 Hz in 10 Hz steps.

CW-only items

Item	Purpose and range
CW speed	5-100 WPM in steps of one.
CW pitch	100-6000 Hz in 10 Hz steps.
Break-in	Turns CW break-in on or off.

10. Spectrum display

When AetherSDR supplies the negotiated spectrum extension, the center area shows a 32-bar audio spectrum with white peak markers. The RX spectrum shows the received audio after DSP, noise reduction, and the RX chain up to and including the equalizer. While transmitting, it shows the TX audio in red across the TX filter width.

During receive, the frequency scale runs from 0 Hz to the active slice's filter width, with a maximum of 12 kHz. The bars are spaced logarithmically by frequency. The lower useful edge is about one-sixtieth of the full span. The axis labels adjust to the current span. Bar height represents a dB value. Peak markers fall gradually.

- Tap the spectrum to show the tiles.
- Swipe left or right to switch between spectrum and tiles.
- If the tiles have not been used for 8 seconds and a stable spectrum is

available, the spectrum returns. It does not return automatically while a tile parameter is assigned to the dial.

- If no new spectrum frame arrives for 5 seconds, the display changes to the

tiles.

- A new stream must remain stable for 1.5 seconds before the display switches

back automatically.

- If AetherSDR supplies no spectrum extension data, the tiles remain visible.

Spectrum generation and handling are implemented and offline-tested. The real spectrum path through AetherSDR and USB/CTR2 has not yet been hardware-verified. A standard CTR2 does not provide this extension.

11. Transmit-related behavior

Each displayed slice has a TX label:

Appearance	Meaning
Gray outline	This is not the transmit slice. Tap it to select this slice for transmit.
Red outline	This is the selected transmit slice, but the radio is not currently transmitting.
Solid red with white text	The radio reports that it is transmitting on this TX slice.

Only an existing slice belonging to the bound GUI station can be selected as the TX slice from this display. The TX label reports and selects the TX slice; it is not a PTT button.

The menu never keys the transmitter. It has no VOX switch and no command that starts TUNE. Tune power changes only the stored TUNE power setting. Use the normal PTT, VOX, or TUNE control outside this menu to start transmitting.

12. Status displays and common states

State	What you see	What to do
No USB host	**Plug into the PC running AetherSDR**	Check the USB cable and PC connection.
Waiting for AetherSDR	**Waiting for AetherSDR**	Set CTR2 Proxy to USB/AetherKnob and press Start.
Radio but no station	**Radio connected … Waiting for a … station**	Connect or select a GUI station in AetherSDR or SmartSDR.
Connected	The status overlay disappears and the round meter face is visible.	The controller is ready.
Locked	Yellow **LOCKED** text	Hold the dial or turn off Tune lock in the menu.
Transmitting	TX label is solid red; spectrum bars are red when TX spectrum data is available.	Status only; it does not key the transmitter.
No spectrum	Tiles remain visible.	Radio control can still work normally. Check spectrum extension support if needed.

13. Troubleshooting

The PC does not detect the device

1. Use the USB-5V-IN connector and try a direct PC connection.
2. Check whether the display or backlight starts. If the device appears dead, check the power source and cable.
3. Press RESET once and wait for USB detection again.
4. If the PC shows **USB JTAG/serial debug unit**, follow the bootloader steps below.

CTR2 Proxy does not list AetherKnob

- AetherSDR must support the AETHER_KNOB USB product string.
- Select Mode: USB, then reopen or refresh the device list.
- Close other programs that may have opened the HID device. Unplug and reconnect AetherKnob, then restart CTR2 Proxy.
- Check that the board is not still running as the ROM USB serial/JTAG device.

The display stays on “Waiting for AetherSDR”

- Open CTR2 Proxy, select USB mode and AetherKnob, then press Start.
- Stop and restart the proxy link, or safely reconnect AetherKnob.
- If the display changes to **Radio connected**, the USB/CTR2 path is working.

The remaining problem is that no AetherSDR or SmartSDR GUI station has been bound.

Touch direction is wrong

The firmware currently uses the touch controller’s X and Y coordinates without a user calibration or rotation setting. Wrong orientation is a known pending hardware check and cannot be corrected in the device menu. Record which physical directions produce which screen movements before changing the firmware configuration.

The rotary dial counts incorrectly or runs backward

The current setting uses four quadrature edges per mechanical detent and reverses the hardware count direction. These settings are not yet hardware-verified and cannot be changed from the device menu. Check whether one detent produces exactly one step and whether clockwise rotation increases frequency or value. A mismatch is a firmware/board calibration issue, not a radio fault.

The device remains in the bootloader or USB JTAG/serial mode

- Release BOOT, then press RESET once.
- If needed, run

python -m esptool --port <COM-port> --after watchdog-reset chip-id, then press RESET again.

- In normal operation, the device should reappear as

AetherSDR AETHER_KNOB, not **USB JTAG/serial debug unit**.

The spectrum is missing

- Radio control may still work. The controller intentionally shows tiles when

no current spectrum data is available.

- A sideways swipe can select the spectrum only while current spectrum data

exists.

- AetherSDR and CTR2 Proxy must negotiate the AetherKnob spectrum extension.

A standard CTR2 does not supply it.

- If no spectrum appears after several seconds, test the other controls and

treat the still-unverified extension/USB data path as a possible cause.

There is no radio control

- Note whether the display says **Waiting for AetherSDR**, **Radio connected**, or

shows the full round meter face.

- Make sure a suitable GUI station and at least one slice exist in AetherSDR.

AetherKnob controls only slices belonging to the bound station.

- **LOCKED** blocks frequency tuning only. Other controls should still work.
- Stop CTR2 Proxy, reset AetherKnob or disconnect USB once, then start the

proxy again with USB/AetherKnob selected.

Safe reset and reconnect procedure

1. If the radio is transmitting, stop transmission with the normal PTT, VOX,

or TUNE control.

2. Stop CTR2 Proxy.
3. Press RESET once. If necessary, unplug and reconnect USB.
4. Wait for **Waiting for AetherSDR**.
5. Start CTR2 Proxy again with USB/AetherKnob selected.

14. Firmware update and flashing

You need an ESP-IDF environment version 6.0 or later. The repository does not currently declare a stable release version.

1. Connect AetherKnob to the PC by USB.
2. Hold BOOT.
3. Press and release RESET.
4. Release BOOT. The ESP32-S3 ROM should now appear as a USB serial/JTAG

device and provide a COM port on Windows.

5. Find the new COM port.
6. Open an ESP-IDF command prompt in the project directory, then build and

flash:

``**powershell idf.py build idf.py -p COMx flash** ``

Replace **COMx** with the actual port.

7. After flashing completes, press RESET once.

8. Allow about 3 seconds for USB to reconnect. The expected device is

AetherSDR AETHER_KNOB, with USB product name AETHER_KNOB.

If the device remains listed as **USB JTAG/serial debug unit**, it is still in download mode. Release BOOT and press RESET again. If required, use the **esptool** command from the troubleshooting section. Runtime logs are then available from the UART connector at 115200 baud, not from the native USB HID connection.

15. Known limitations

• Display startup on the real CrowPanel is not yet recorded as verified.

• Touch orientation and gestures are not yet confirmed on the target device.

There is no user touch calibration in the menu.

• Encoder direction and four counts per detent are not yet calibrated on the

target device.

• Meter data through the relayed UDP path, including live meter updates, is not

yet hardware-verified.

• The spectrum path is implemented and offline-tested, but the real

AetherSDR/USB path is not yet hardware-verified.

• The controller depends on AetherSDR CTR2 Proxy. It has no direct network

connection to the radio.

• NRL, NRS, RNN, NRF, ANFL, and ANFT are intended for FLEX-8000/Aurora radios.

A FLEX-6000 ignores these commands.

• The menu does not key the transmitter. VOX and starting TUNE are deliberately

absent.

• The repository does not declare a stable release version.

16. Quick reference

Action	Result
Turn the dial	Change frequency by the current tuning step.
Short-click the dial	Select the next tuning step; in parameter mode, return to tuning.
Hold the dial for at least 0.6 s	Toggle Tune lock; in the menu, go back or close the menu.
Tap a tile	Assign its parameter to the dial; tap again to leave parameter mode.
Hold NR, NB, or ANF	Turn that function on or off.
Hold AF or AGC	Step the AGC mode.
Hold FILT	Select the next tuning step.
Tap a TX label	Select that slice as the TX slice.
Swipe down	Open the menu.
Swipe left or right	Switch between spectrum and tiles when spectrum data is available.
Tap the spectrum	Show the tiles.
Swipe the menu up or down	Scroll the list.
White menu outline	Row is highlighted; the dial moves through the menu.
Yellow menu outline	Row is being edited with the dial.
Tap or swipe up from the bottom grip	Close the menu.

17. First-use checklist

- [ ] Switch on the FlexRadio.
- [ ] Start AetherSDR and connect it to the required station.
- [ ] Connect AetherKnob to the PC through USB-5V-IN.
- [ ] Wait for **Waiting for AetherSDR**.
- [ ] Open CTR2 Proxy in AetherSDR.
- [ ] Select Mode: USB.
- [ ] Select AetherKnob / AETHER_KNOB.
- [ ] Press Start.
- [ ] Wait until the status overlay disappears and the round meter face appears.
- [ ] Check that the required active slice is shown at the top.
- [ ] Turn the dial carefully by one detent and confirm one tuning step.
- [ ] If the wrong slice is active, select it in AetherSDR first. Tap a TX label

only when you also intend to change the transmit slice.